Three survival strategies: a viability taxonomy of cosmological scenarios

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Abstract

The viability study of self-consuming systems identified three survival strategies — *residence* in a dimensionless processing window, *periodic transit* through it, and *dilution* of damage by growth — as the regions of a two-parameter diagram (x, γ). We propose that these three regions classify cosmological scenarios, and that *the celebrated theoretical pathology of each family is exactly the death predicted for its region.* Under a pre-registered protocol (dictionary and kill conditions fixed before evaluation), we non-dimensionalize each family: inflation lives in the dilution region with $D \equiv N_{\mathrm{supplied}}/N_{\mathrm{required}}$ ($N_* = 50$–60, sourced; hilltop models sit at the threshold, large-field models far above it), and its graceful-exit problem is the dilution trap; cyclic models live in transit with $R \equiv S_{\mathrm{erased}}/S_{\mathrm{produced}}$ per cycle (Tolman's lengthening cycles are $R = 0$; conformal cyclic cosmology claims $R \to 1$ conditionally on complete black-hole evaporation — contested, declared; the ekpyrotic scenario is *reclassified* as a transit–dilution hybrid: it dilutes its debt rather than repaying it); sustained phantom cosmologies die the autotic death in its *exact* form — any $x = -w > 1$ maintained eternally diverges in finite time ($t_{\mathrm{rip}} \simeq 231$ Gyr at $w = -1.05$), making cosmology the exact-theorem version of the two-tier structure the biology only measures; and eternal Λ (with Hoyle's steady state, the two cosmologies frozen at exact criticality $x = 1$) receives its own celebrated death, the recurrence/Boltzmann-brain pathology. A landscape multiverse is not an orphan but a *population*, excluded by the same guard that excluded the honeybee colony and handed to demography (companion paper B): the two frameworks partition the world without overlap, by their guards alone. All three kill conditions — orphan scenarios, pathologies in viable regions, region flips under alternative damage definitions — were faced after non-dimensionalization; none fired. An epistemic control bounds the claim: Hoyle's steady state was strategy-viable and observationally false — the taxonomy classifies ways of surviving, never truth.

1 Claim and protocol

Companion work established a dimensionless viability window for self-consuming systems ($x = \tau_c/\tau_s$; for a fed universe, exactly $x = -w$) and a two-parameter extension (x, γ) whose three regions are the observed survival strategies. Here we test one claim: *cosmological scenarios partition into these regions, with each family's classic pathology being its region's predicted death.* The dictionary was fixed before evaluation: damage = the problem each scenario's *own* literature says it

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solves or suffers (never "whatever the mechanism removes" — the tautology guard); processing = feeding/thermalization; dilution = e-folds carrying damage away; transit = a bounce with an explicit debt-clearing mechanism. Kill conditions, pre-registered: (M1) a major scenario fits no region or needs a fourth; (M2) a known pathology sits comfortably in a viable region; (M3) a defensible alternative damage definition flips any scenario's region.

2 Non-dimensionalization

Family	Region	Dimensionless	Its celebrated pathology = it
Inflation	dilution	$D = N/N_*$, $N_* = 50$–60 (hilltop $D \simeq 1$; large-field $D \gg 1$)	graceful exit = the dilution t (eternal: $D \to \infty$, never leav
Reheating	dilution→residence (the handover)	thermal entropy $\sim 10^{89}k$ (not the all-time max: SMBHs hold $10^{104}k$ [sourced])	the cosmic Mandelstam test: dilution stops, processing mu — and it does
Cyclic (Tolman)	transit	$R = S_{\text{erased}}/S_{\text{produced}} = 0$	lengthening cycles = unpaid
CCC (Penrose)	transit	$R \to 1$ *conditional* on total BH evaporation [contested, declared]	the condition is the repayme mechanism itself
Ekpyrotic	**transit–dilution hybrid**	S_{tot} grows, density $s \to 0$	reclassified: dilutes its debt
Sustained phantom	supercritical residence	margin $(x-1) > 0$ eternal	**exact autosis**: t_{rip} finite
Fed era (A/B)	residence	$x(t) = -w(t)$: 1.35 → 0.85 → exit at 16–36 Gyr if feast finite	neuron's death at feast end ($t_{\text{end}}/t_0 = (M_{\text{res}}/M_{\text{acc}})^{1/\beta}$)
Λ eternal; Hoyle	frozen criticality	$x = 1$ exactly, forever (Hoyle: by construction)	recurrences / Boltzmann bra [10]; Hoyle: the epistemic co
Emergent (static seed)	knife-edge residence	fixed point	Einstein-static instability
Landscape multiverse	*excluded by guard*	a population, not a system	handed to demography (pap

Table 1: The taxonomy after non-dimensionalization. Sourcing: N_* [1, 2]; entropy budget [3]; $t_{\text{rip}} = \frac{2}{3}/(|1+w|H_0\sqrt{\Omega_\Lambda})$ [computed: 231 Gyr at $w = -1.05$]; Tolman [5]; ekpyrotic [7]; CCC [6]; Big Rip [8]; Hoyle [9]; feast-end [computed, companion audit].

The exact theorem. The phantom row upgrades the two-tier duration structure from measurement to theorem. In biology, sustainable ceilings exceed transient ones by measured factors (7× for days vs. 2.5× indefinitely). In cosmology the statement is exact: transient supercriticality ($x > 1$ for finite eras — our own past phantom phase) is survivable; *eternal* supercriticality diverges in finite time, however mild ($x = 1.05 \Rightarrow 231$ Gyr). Criticality itself is the sustained edge, with no free parameter. Conversely, frozen criticality forever ($x = 1$: Λ, Hoyle) has its own celebrated pathology [10] — M2 holds even where immortality was assumed.

3 Kill conditions faced

M3 (alternative damage). Inflation: horizon ($N_* \approx 60$) vs. curvature (≈ 50–60) vs. monopoles (≈ 23–60, scale-dependent): D shifts by $< 3\times$, no region change; and "damage = self-generated fluctuations" does not move healthy inflation — it *explains* the pathological subcase (self-reproduction is where damage creation beats dilution). Cyclic: total entropy vs. entropy density vs. surviving black holes changes the *required repayment mechanism* (ekpyrotic dilutes; CCC must evaporate), never the region. Phantom: the death is geometric, damage-definition-free. **No flips: M3 passes.**
M1 (orphan hunt). The landscape multiverse is a population — the guard excludes it exactly as

it excluded the honeybee colony, and demography (paper B) is its proper science: the two frameworks partition by their guards, with no overlap and no orphan. Emergent universes are knife-edge residence (their pathology: the Einstein-static instability); single bounces are one-shot transit. No orphan among $\sim$12 major scenarios — with the declared caveat that only two hunters searched. *M2*: every examined pathology landed on its region's death, including Λ's.

4 What this is, and is not

The taxonomy classifies *ways of surviving*, never truth: Hoyle's steady state — perfect criticality by construction — was strategy-viable and killed by the CMB. Contested and declared: CCC's R; the model-class distribution of D; and the structural risk that rhymes between confident minds are overproduced — external review matters here more than anywhere in the series. What would kill it: an orphan scenario; a pathology at home in a viable region; a region flip under a defensible damage definition; or a demonstration that the mapping's dictionary smuggles in its conclusions. One resonance is recorded without weight: since reheating, the universe's dominant entropy production has been the manufacture of black holes [3, 4] — the parent-making of companion paper B; whether that is bookkeeping or teleology is precisely the kind of question this taxonomy is built not to answer.

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